

Lab Practical 01

Software Engineering Framework Fundamentals, Framework Selection & Emerging Trends

Lab Practical Number	Lab 01
Lab Title	Software Engineering Framework Fundamentals, Framework Selection & Emerging Trends
Related Lecture	Lecture 01 – Software Engineering Framework Fundamentals, Selection and Emerging Trends
Duration	2 Hours (in-lab)
Learning Outcomes	LO1 – Evaluate frameworks • LO4 – Select and justify frameworks, tools & AI-assisted approaches
Mode	Individual work • Word document submitted to CourseWeb within the session

Introduction

This lab puts Lecture 01 into practice. In the lecture you learned what a software engineering framework is, how a framework differs from a library, the main framework families used across web, mobile, cloud, and AI-assisted development, and a repeatable method for selecting and justifying frameworks for a real project. In this 2-hour session you will apply that knowledge to concrete classification, comparison, selection, and trend-analysis tasks.

Everything you need is already in Lecture 01. Base your answers on the lecture content, the worked CampusEats case study, and the class discussion. You may add suitable examples of your own, but stay within the scope of Lecture 01 — do not introduce unrelated topics.

Lab Objectives

By the end of this lab, you should be able to:

- **Distinguish frameworks from libraries** using the control-of-flow (Inversion of Control) test, and justify each classification.
- **Compare frameworks within a family** (frontend, backend, or mobile) against criteria such as language, best fit, strengths, and industry adopters (LO1).
- **Apply a selection workflow** and a weighted decision matrix to choose a suitable framework for a given project context (LO4).
- **Plan a full-stack technology stack** for a real-world scenario and justify every layer against an explicit selection criterion (LO4).
- **Analyse emerging trends** — microservices, serverless, cloud-native/DevOps, and AI-assisted/agentive development — and judge when each is appropriate (LO1, LO4).

Required Tools / Materials

- **Lecture 01 slide deck** — the primary reference for all tasks (definitions, comparison tables, selection criteria, case study, trends).

- **A word processor** — Microsoft Word (or Google Docs / LibreOffice exported to .docx) to prepare the submission.
- **A web browser** — for quick reference to official framework documentation (react.dev, learn.microsoft.com/aspnet/core, docs.flutter.dev) — optional.
- **This lab sheet** — for task instructions and the quick test.

Note: no coding or installation is required for this lab. The focus is evaluation, comparison, selection, and justification — the skills behind LO1 and LO4.

Lab Time Plan (120 minutes)

Time	Activity	Output
0:00 – 0:10	Briefing & recap of Lecture 01 key ideas (framework, IoC, families, selection method)	Ready to start
0:10 – 0:30	Task 01 — Framework or Library? Classify and justify	Classification table
0:30 – 0:55	Task 02 — Build a framework comparison table for one family	Comparison table + verdict
0:55 – 1:25	Task 03 — Selection workflow, weighted matrix & stack plan	Matrix + stack + sketch
1:25 – 1:50	Task 04 — Real-world case selection + emerging-trends reflection	Stack + trend analysis
1:50 – 2:00	Quick test + finalise and submit the Word document	Submitted .docx

HOW TO USE THIS PLAN

The times are a guide. Aim to keep Tasks 01–02 brisk so you have enough time for the selection and trends work in Tasks 03–04, which carry the most marks.

TASK 01 • 20 minutes • LO1

Framework or Library? Classify and Justify

Recall the decisive test from the lecture: **who controls the flow of execution?** If *your* code calls the tool whenever you choose, it is a **library**. If the tool runs the main loop and calls *your* code at its extension points (Inversion of Control — the “Hollywood Principle”), it is a **framework**.

Step-by-step instructions

1. Copy the table below into your Word document.
2. For each technology, write Framework or Library in the second column.
3. In the third column, give a one-line justification using the control-of-flow test (state who calls whom).
4. For the item marked with a star (★), explain why it sits in the “grey zone” discussed in the lecture and defend whichever classification you choose.

Technology	Framework / Library	Justification (who controls the flow?)
lodash		
ASP.NET Core		
axios		
Flutter		
jQuery		
Angular		
Express.js		
NumPy		
React ★		
Spring Boot		

EXPECTED OUTPUT

A completed 10-row table with a correct classification and a one-line, control-flow-based justification for each technology, plus a short paragraph (2–3 sentences) defending your classification of React.

TASK 02 • 25 minutes • LO1

Build a Framework Comparison Table

In the lecture we surveyed six framework families. In this task you compare frameworks **within one family** so you can evaluate options the way a professional does — on evidence, not popularity.

Step-by-step instructions

1. Choose **one** framework family from the list below.

Family	Frameworks to compare (pick from these)
Frontend (Web UI)	React • Angular • Vue
Backend / Server-side	ASP.NET Core • Spring Boot • Express.js / NestJS • Django / FastAPI
Mobile	Flutter • React Native • Native (SwiftUI / Jetpack Compose)

2. Compare at least two (ideally three) frameworks from that family by completing the comparison table below.
3. Fill every cell using Lecture 01 content; you may add one fact of your own per framework if clearly relevant.
4. Below the table, write a short verdict (3–4 sentences): for a **3-developer student startup building its first product in 4 months**, which framework in your family would you start with, and why?

Criterion	Framework A: ____	Framework B: ____	Framework C: ____
Type			
Language			
Best fit / strengths			
Used by (industry)			
Main trade-off / cost			

EXPECTED OUTPUT

A completed comparison table for one framework family (2–3 frameworks across 5 criteria) plus a short, justified verdict naming your recommended starting framework for the given startup context.

TASK 03 • 30 minutes • LO4

Selection Workflow, Weighted Matrix & Stack Plan

Now apply the professional selection method from the lecture: **Requirements → Constraints → Shortlist → Prototype → Decide & record**. You will score candidates objectively with a weighted decision matrix, then plan a full stack — citing a criterion for every choice, exactly as in the CampusEats case study.

Scenario

“GREENCART” — CAMPUS GROCERY PICKUP

A team of **3 developers** has **4 months** and a small budget to build GreenCart: a service that lets students order groceries from campus shops and collect them from a pickup point.

Requirements: a student-facing mobile app (iOS + Android), a shop-owner web dashboard, order management and simple payments, and the system must be able to scale if it becomes popular across faculties.

Step-by-step instructions

- Part A — Weighted matrix.** Pick **one** layer to evaluate in detail (frontend *or* mobile *or* backend). Complete the matrix below: assign each criterion a weight (the weights must total 100%), score each candidate 1–5, then compute the weighted total. State which candidate wins.
- Part B — Stack plan.** Decide a framework/technology for every layer of GreenCart and complete the stack table. Each justification must cite **one selection criterion** from the lecture (e.g. project fit, team skills, community & ecosystem, performance, maturity, security, cost, integration).
- Part C — Architecture sketch.** Draw a simple block diagram of your stack (mobile app + web dashboard → backend API → database, with the delivery/hosting layer noted). Hand-draw and photograph it, or build it with shapes/SmartArt in Word.

Part A — Weighted decision matrix

Criterion	Weight %	Candidate 1	Candidate 2	Candidate 3
Team familiarity				
Time to first working version				
Hiring pool (if team grows)				
Ecosystem for our features				
Long-term maintainability				
Weighted total	100%			

Winner of this layer: _____ (the highest weighted total)

Part B — GreenCart technology stack plan

Layer	Your choice	Justification (cite one criterion)
Mobile app		
Web dashboard		
Backend API		
Database		

Layer	Your choice	Justification (cite one criterion)
Delivery & hosting		

EXPECTED OUTPUT

(A) A completed weighted matrix for one layer with weights summing to 100%, scores, weighted totals, and a named winner. (B) A five-layer stack table where every choice cites a selection criterion. (C) A simple architecture sketch of the GreenCart system.

ASSESSOR HINT FOR YOU

Strong answers read like the CampusEats case study: every choice traces to a criterion, not a preference. “Flutter, because one codebase covers iOS + Android and a 3-person team cannot staff two native tracks” is a justification; “Flutter, because it’s popular” is not.

TASK 04 • 25 minutes • LO1 & LO4

Real-World Case + Emerging-Trends Reflection

Frameworks change every few years; the *selection method* and an awareness of *emerging trends* are what last. In this task you act as the tech lead for a fresh real-world system, choose a stack, then judge where modern architectural trends do — and do not — fit.

Choose ONE case scenario

Option	Scenario
A	A private hospital chain wants a patient appointment-booking system. Security and compliance are critical; staff use a web portal and patients use a mobile app.
B	A nationwide bus company wants a real-time seat-booking and live-tracking platform with heavy, spiky traffic during holiday seasons.
C	A clothing brand wants a mobile-first e-commerce app that must scale sharply during seasonal sales and promotions.

Part A — Select and justify a stack (LO4)

1. State which scenario you chose and list its two most important non-functional requirements (e.g. security, scalability, time-to-market).
2. Choose a framework/technology for frontend (or mobile), backend, database, and hosting.
3. Name your **top three selection criteria** for this scenario and explain *why* each matters here. Your stack choices must reflect those criteria.

Part B — Emerging-trends reflection (LO1)

For the same system, complete the table: for each trend, say whether you would apply it, give a one-line justification, and name one risk or cost. “Not now” is a perfectly valid, well-reasoned answer (recall: newer is not automatically better).

Trend	Apply? (Yes/No)	Where / why	One risk or cost
Microservices			
Serverless functions			
Cloud-native & DevOps (CI/CD)			
AI-assisted / agentic development			

EXPECTED OUTPUT

(A) A justified stack for your chosen scenario, led by your top three criteria. (B) A completed four-row trends table with a reasoned Apply/Not-now decision and a stated risk for each trend. One sentence reflecting on the lecture’s key insight: the options change, the method does not.

QUICK TEST • 10 minutes**Knowledge Check**

Answer all questions in your Word document. Write the question number and your answer. Questions 1–5 are multiple choice; 6–7 are short answer; 8 is scenario-based.

Multiple choice (choose one)

Q1. The defining difference between a framework and a library is:

- (a) size
- (b) programming language
- (c) who controls the flow of execution
- (d) the licence

Q2. “Don’t call us, we’ll call you” (the Hollywood Principle) describes:

- (a) REST
- (b) Inversion of Control
- (c) CI/CD
- (d) serverless computing

Q3. A 3-person startup needs iOS + Android apps in 4 months. The strongest first candidate is:

- (a) two separate native apps
- (b) Flutter (cross-platform)
- (c) a desktop application
- (d) a static website

Q4. The best reason to choose framework X over framework Y is that X:

- (a) is trending on social media
- (b) is used by Google
- (c) has the highest weighted score against your project’s criteria
- (d) had the newest release this week

Q5. In agentic / AI-assisted development, the human engineer’s essential role is to:

- (a) type faster
- (b) review, test and take responsibility for the output
- (c) avoid writing documentation
- (d) remove all human oversight

Short answer (1–2 sentences each)

Q6. Give one benefit AND one cost of adopting a large framework for a short student project.

Q7. Name the five stages of the framework selection workflow taught in the lecture, in order.

Scenario-based

Q8. A classmate says: “I imported axios into my project, so now I’m using a framework.” Is this correct? Justify your answer in one or two sentences using the control-of-flow test.

SUBMIT YOUR QUICK-TEST ANSWERS INSIDE THE SAME WORD DOCUMENT

Place your quick-test answers in a clearly labelled “Quick Test” section after your Task 01–04 answers.

Submission Instructions

Prepare a single Microsoft Word document containing all of your work for this lab.

Your Word document must include:

- Student name
- IT number
- Lab practical number (Lab 01)
- Your answers for Task 01, Task 02, Task 03 and Task 04
- Your Quick Test answers (Q1–Q8)

File naming and upload:

1. Rename the Word document using your IT number in this exact format: **ITXXXXXXXX_Lab01.docx**
2. Example: **IT22123456_Lab01.docx**
3. Upload the completed document to **CourseWeb** before the end of the 2-hour lab session.
4. **Late submissions will not be accepted** unless prior approval is given by the lecturer.

STUDENT INSTRUCTION

You must refer to the provided Lecture 01 material when answering the lab tasks. Your answers should be based on the lecture content, class explanations, and the practical examples discussed in the lecture.

Evaluation Criteria (Marking Guide — 20 Marks)

For lecturer reference. Marks reward correct classification, evidence-based comparison, and — above all — justification that traces to explicit criteria.

Component	Marks	What earns full marks
Task 01 — Framework vs Library	4	All 10 classified correctly; each justified by the control-of-flow test; React grey-zone defended.
Task 02 — Comparison table	4	Table complete for 2–3 frameworks across all criteria; verdict is justified for the given startup context.
Task 03 — Selection, matrix & stack	5	Weighted matrix sums to 100% with computed totals and a named winner; full stack with a criterion cited per layer; architecture sketch present.
Task 04 — Case + trends reflection	4	Stack led by top-3 criteria; trends table reasoned with risks; sound “apply / not now” judgement.
Quick Test (Q1–Q8)	3	Q1–Q5 correct (c, b, b, c, b); Q6–Q8 show correct understanding.
Total	20	